MINI PROJECT PORTFOLIO

2D GRAPHICS EDITOR USING C

Submitted By

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ABSTRACT

The 2D Graphics Editor is a menu-driven application developed using the C programming language. The project provides a text-based graphical environment where users can create and manipulate basic geometric shapes such as rectangles, lines, triangles, and circles on a 20×40 canvas. The editor supports object creation, deletion, modification, display, and canvas reset operations.

OBJECTIVES

1. Create a 2D drawing canvas using a character array.
2. Implement drawing of geometric shapes.
3. Support deletion and modification of objects.
4. Display graphical output in the console.
5. Apply modular programming concepts in C.

PROBLEM STATEMENT

Design and implement a menu-driven 2D Graphics Editor using the C programming language.

SYSTEM REQUIREMENTS

Hardware: Intel Core i3+, 4 GB RAM.
Software: Windows 10/11, C, Code::Blocks, GitHub.

SYSTEM DESIGN

Canvas Size: 20 rows × 40 columns. Coordinate range x:0-39, y:0-19. The top-left corner is (0,0).

DATA STRUCTURES USED

Object Types: Rectangle, Line, Triangle, Circle.
Canvas: 20×40 character array.
Object Storage: Array of objects.

FEATURES IMPLEMENTED

Canvas Initialisation, Display Canvas, Draw Rectangle, Draw Line, Draw Triangle, Draw Circle, Delete Object, Modify Object, Reset Canvas, Exit Application.

ALGORITHM

Initialise canvas, display menu, accept user choice, perform operation, render objects, display canvas, repeat until exit.

IMPLEMENTATION DETAILS

Rectangle, Line, Triangle, and Circle drawing implemented using coordinates and dimensions. Circle uses (x-cx)^2 + (y-cy)^2 <= r^2.

TEST CASES

Rectangle(5,5,10,5), Line(2,2,8), Triangle(20,10,6), Circle(30,10,4), Delete Object, Modify Object.

OUTPUT SCREENSHOTS

Insert screenshots of menu, rectangle, line, triangle, circle, delete, modify, and reset operations.

```
--- Main Menu ---
1. Draw Rectangle
2. Draw Line
3. Draw Triangle
4. Draw Circle
5. Delete Object
6. Modify Object (Move)
7. Display Canvas
8. Reset Canvas
9. Exit
Select an option (1-9):
```

```
Rectangle added successfully!

   0123456789012345678901234567890123456789
 0 ________________________________________
 1 ________________________________________
 2 ________________________________________
 3 ________________________________________
 4 ________________________________________
 5 _____**********_________________________
 6 _____*________*_________________________
 7 _____*________*_________________________
 8 _____*________*_________________________
 9 _____**********_________________________
10 ________________________________________
11 ________________________________________
12 ________________________________________
13 ________________________________________
14 ________________________________________
15 ________________________________________
16 ________________________________________
17 ________________________________________
18 ________________________________________
19 ________________________________________
```

```
Horizontal Line added successfully!

   0123456789012345678901234567890123456789
 0 ________________________________________
 1 ________________________________________
 2 __********______________________________
 3 ________________________________________
 4 ________________________________________
 5 ________________________________________
 6 ________________________________________
 7 ________________________________________
 8 ________________________________________
 9 ________________________________________
10 ________________________________________
11 ________________________________________
12 ________________________________________
13 ________________________________________
14 ________________________________________
15 ________________________________________
16 ________________________________________
17 ________________________________________
18 ________________________________________
19 ________________________________________
```

```
Right-angled Triangle added successfully!

   0123456789012345678901234567890123456789
 0 ________________________________________
 1 ________________________________________
 2 __********______________________________
 3 ________________________________________
 4 ________________________________________
 5 ________________________________________
 6 ________________________________________
 7 ________________________________________
 8 ________________________________________
 9 ________________________________________
10 ____________________*___________________
11 ____________________**__________________
12 ____________________***_________________
13 ____________________****________________
14 ____________________*****_______________
15 ____________________******______________
16 ________________________________________
17 ________________________________________
18 ________________________________________
19 ________________________________________
```

```
Circle added successfully!

   0123456789012345678901234567890123456789
 0 ________________________________________
 1 ________________________________________
 2 __********______________________________
 3 ________________________________________
 4 ________________________________________
 5 ________________________________________
 6 ______________________________*_________
 7 ____________________________*****_______
 8 ___________________________*******______
 9 ___________________________*******______
10 ____________________*_____*********_____
11 ____________________**_____*******______
12 ____________________***____*******______
13 ____________________****____*****_______
14 ____________________*****_____*_________
15 ____________________******______________
16 ________________________________________
17 ________________________________________
18 ________________________________________
19 ________________________________________
```

```
Success: Deleted Object ID 5 (Type 4) found at (30, 10).

   0123456789012345678901234567890123456789
 0 ________________________________________
 1 ________________________________________
 2 __********______________________________
 3 ________________________________________
 4 ________________________________________
 5 ________________________________________
 6 ________________________________________
 7 ________________________________________
 8 ________________________________________
 9 ________________________________________
10 ____________________*___________________
11 ____________________**__________________
12 ____________________***_________________
13 ____________________****________________
14 ____________________*****_______________
15 ____________________******______________
16 ________________________________________
17 ________________________________________
18 ________________________________________
19 ________________________________________
```

```
Canvas and all objects have been reset.

   0123456789012345678901234567890123456789
 0 ________________________________________
 1 ________________________________________
 2 ________________________________________
 3 ________________________________________
 4 ________________________________________
 5 ________________________________________
 6 ________________________________________
 7 ________________________________________
 8 ________________________________________
 9 ________________________________________
10 ________________________________________
11 ________________________________________
12 ________________________________________
13 ________________________________________
14 ________________________________________
15 ________________________________________
16 ________________________________________
17 ________________________________________
18 ________________________________________
19 ________________________________________
```

ADVANTAGES

Simple interface, demonstrates C concepts, modular design, supports multiple objects.

FUTURE ENHANCEMENTS

Colored objects, filled shapes, save/load drawings, GUI implementation.

CONCLUSION

The project successfully implements a 2D Graphics Editor using C with object creation, modification, deletion, and display capabilities.

REFERENCES

Kernighan & Ritchie - The C Programming Language;

Code::Blocks Documentation;

GCC Documentation;

GitHub Documentation.